

Towards a Multi-Agent Vision-Language System for Zero-Shot Novel Hazardous Object Detection for Autonomous Driving Safety

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Abstract—Detecting anomalous hazards in visual data, particularly in video streams, is a critical challenge in autonomous driving. Existing models often struggle with unpredictable, out-of-label hazards due to their reliance on predefined object categories. In this paper, we propose a multimodal approach that integrates vision-language reasoning with zero-shot object detection to improve hazard identification within traffic scene data. Our approach consists of a pipeline that integrates both Vision-Language Model (VLM) and Large-Language Model (LLM) capabilities to solve the problem of detecting out-of-label hazards within traffic scene data. We refine object detection by incorporating OpenAI’s CLIP model to match predicted hazards with bounding box annotations, improving localization accuracy. To assess model performance, we extended the foundational COOOL (Challenge-of-Out-of-Label) anomaly detection benchmark dataset by adding full natural language hazard descriptions as ground truth annotations. Additionally, we evaluated our pipeline’s hazard labels through cosine similarity in order to consider the semantic similarity between the predicted hazard description and the annotated ground truth for each video. Our findings highlight the strengths and limitations of current vision-language based approaches, offering insights into future improvements in autonomous hazard detection systems. Our models, scripts, and data can be found at <https://github.com/mi3labucm/COOLER.git>

Index Terms—hazard recognition, novelty detection, video understanding, autonomous driving

I. INTRODUCTION

The rise of autonomous vehicles underscores the challenge of ensuring safety in diverse and unpredictable environments. While significant progress has been made in handling standard road conditions, autonomous systems often struggle with rare or unconventional conditions and hazards that fall outside typical training distributions [1]–[6]. Addressing these limitations is crucial for enhancing the robustness and real-world deployment of autonomous driving technology.

This research advances out-of-label hazard detection by integrating Vision-Language Models (VLMs) such as ViLA [7], [8], OmniVLM [9] and OpenAI’s GPT-4o-mini vision capabilities, alongside a Large Language Model (LLM) through OpenAI’s GPT-4o-mini, to identify rare and unconventional traffic hazards. We then incorporate OpenAI’s CLIP model in order to match predicted hazards with bounding box data

to improve localization accuracy. Unlike traditional datasets that emphasize common driving conditions, our approach targets anomalies, which include debris, animals, and erratic pedestrian behaviors that pose critical but often overlooked risks [10].

Building from the landmark COOOL (Challenge-of-Out-of-Label) benchmark¹ dataset [11], we introduce a dataset extension and performance measurement method to validate our approach to anomaly recognition. This dataset extension includes preprocessing of the COOOL videos, ground truth hazard label annotations, and definition of an open-set method of evaluation to validate our experiments and our methods of anomaly detection and description. The dataset within this benchmark includes 200 short video clips capturing diverse traffic scenes featuring anomalies and hazards and annotated bounding boxes for all objects within the scene. We name our augmentation of the dataset *COOLER* (COOOL with Expanded Representations); our augmentation of the COOOL benchmark includes the following for each video scene:

- 1) Removal of video watermarks for more naturalistic visual driving scenes
- 2) Denoising and deblurring of the traffic scenes for enhanced video quality via NAFNet [12]
- 3) Human annotations for the descriptions of each Hazardous object within the video sequence.
- 4) Publicly available, open-set methods of evaluation to human annotated ground truth hazard descriptions which include cosine similarity and embedding based caption analysis

By bridging vision-language understanding with improved hazard detection, this research enhances the ability of autonomous vehicles to recognize real-world anomalies through the use of VLMs, LLMs, CLIP and a new evaluation bench-

¹To clarify our terminology, we use *benchmark* to refer to a standardized dataset, task, and means of evaluation. While COOOL provides unlabeled data, in its current form, the benchmark does not provide annotated hazards and only allows black-box evaluation based on string subsequence matching to an unknown string set, necessitating an open-set language extension for validating our hazard recognition methods.

mark through COOLER, ultimately improving safety and reliability in unpredictable autonomous driving conditions.

II. RELATED RESEARCH

A. Vision Language Models

In recent years, vision-language-models have made significant advancements in bridging the gap between visual perception and linguistic reasoning [13], [14]. These models, including ViLA [8], OmniVLM [9], GPT-4o-mini [15], and CLIP [16], have demonstrated strong capabilities in tasks such as image captioning, question answering, and zero-shot object recognition [17]. The broader class of vision-language models has improved scene understanding by enhancing the ability to describe, interpret, and analyze visual inputs, making them increasingly relevant in domains like autonomous driving, where accurate perception and contextual reasoning are essential [6], [18], [19]. Beyond these models, advancements in image and video processing have further enhanced the quality of vision-language tasks. However, despite these advancements, significant challenges remain [20], particularly in anomaly detection where models must recognize subtle deviations in dynamic environments. Future developments in VLMs will likely focus on refining their ability to detect rare hazards, improving their robustness in novel situations, and strengthening their capacity for contextual understanding in real-world autonomous systems.

B. Autonomous Driving Benchmarks

The creation of benchmarks such as nuScenes [21], Cityscapes [22], and KITTI [23] have contributed to the development of autonomous driving technologies. These datasets provide high quality annotations for tasks like segmentation, trajectory prediction, and object detection; however, they primarily focus on well defined, common scenarios. For example, the nuScenes dataset consists of 1,000 annotated scenes, lasting 20 seconds each, comprising a total of 1.4 million images and detailed 3D bounding box annotations. These annotations cover various conditions, including low light, rain, and dense traffic, making them highly versatile. However, the nuScenes dataset has limitations in addressing unconventional hazards or rare occurrences. nuScenes and similar datasets such as KITTI and Cityscapes primarily focus on predefined categories that fail to capture the “novelty problem” where autonomous systems encounter anomalous events.

The DHPR (Driving Hazard Prediction and Reasoning) dataset addresses the challenge of predicting traffic hazards and the “novelty problem” by introducing abductive reasoning from static dashcam images [24]. It consists of 15,000 annotated scenes, each paired with car speed, bounding boxes, and natural language hazard explanations. By focusing on high level reasoning over visual cues, DHPR establishes a benchmark for integrating vision-language models into hazard prediction. However, a key limitation is its reliance on single images, which excludes temporal cues essential for modeling dynamic driving contexts. Additionally, the hazards are hypothetical rather than real, derived through human annotation.

These constraints limit DHPR’s capacity to fully represent real-world driving risks, highlighting the need for datasets that integrate temporal sequences and richer contextual data for comprehensive hazard analysis.

C. Existing Approaches to Hazard Detection

Recent work in hazard detection includes multimodal and vision-language based frameworks [20], [25]–[27]. Zhouxiang and Petrosian [28] propose a multimodal driver assistance system that integrates road conditions, facial video analysis, and additional sensory inputs to improve hazard detection. Their approach focuses on fusing multiple data sources to enhance situational awareness. Chen et al. [29] introduces a hierarchical VLM framework to enhance hazard detection by leveraging visual representations and semantic reasoning. Their framework interprets driving scenarios and predicts potential dangers by analyzing scene context. Our work builds upon these efforts by employing a multi-agent vision-language system designed to reason about scene context, identify anomalies, and recognize hazards in previously unseen scenarios. Instead of relying on multimodal sensor fusion, we focus on VLMs such as ViLA, OmniVLM, and GPT-4o-mini to interpret complex visual and textual information, enabling a more adaptable and generalized hazard pipeline. Xiao et al. propose a kinematics-based framework for detecting hazardous objects in near miss traffic scenarios [30] using the GTACrash dataset [31], a simulation generated collection of 7,700 crash and 3,600 near miss events. They compute evasive maneuvers and dynamically determined safe distances by analyzing vehicle deceleration, acceleration, jerk, and longitudinal/lateral gap thresholds. To address inconsistent labeling across datasets where hazard definitions vary they introduce configurable, event agnostic safety metrics that standardize annotations by relying on automatically detected kinematic triggers. However, because their method uses only vehicle motion data and ignores visual context, it cannot capture drivers’ intent or environmental factors, such as occlusions or visual hazards. Additionally, GTACrash’s simulated nature limits realism and diversity compared to real-world conditions. Pinggera et al. [32] introduce a stereo vision-based method to detect small road obstacles, some as small as 5 cm from a distance of 20 meters, addressing a critical challenge in autonomous driving safety. The researchers presented the Lost and Found dataset, a collection of over 2,000 annotated stereo frames tailored for hazard detection in diverse road scenarios. The stereo vision-based detection approach achieves a high accuracy score in detecting small hazards with low false positive rates, showing promising results in comparison to other stereo-based approaches. However, its reliance on stereo vision restricts accuracy in long range or low texture areas and the dataset lacks temporal reasoning. This is because it focuses only on static frames rather than dynamic hazards. Importantly, our approach expands beyond this method by using vision-language reasoning for more complete contextual understanding and description.

Abbarkiki et al. uses a real-time pipeline that combines depth estimation, optical flows, and VLMs [33]. Through the

use of DPT-DINOv2 for monocular depth, RAFT for optical flow, bounding box data identifying hazardous areas, and the BLIP-2 vision-language model, they are able to generate hazard explanations. However, each frame in the pipeline is sent separately through both DPT-DINOv2 and RAFT which could lead to errors in either of the networks propagating out. This process also requires a minimum amount of history on specific conditions before flagging certain events as hazards. This leads to this pipeline being delayed when trying to detect sudden hazardous events.

Hatami et al. introduce a stacked process that uses NAFNet for image restoration along with MiDaS for depth-based filtering and SegFormer-b5 for the segmentation of drivable area [34]. However, their method depends on matching captions to a fixed set of heuristic rules for hazard classification, limiting the coverage of real-world changes. Furthermore, once assigned, these labels persist on the frame for the entire duration, preventing the system from recognizing when the hazard has ended.

We expand on the above research ideas by introducing a new approach to out-of-label hazard detection in traffic scenarios by implementing a multi-agent system that leverages zero-shot vision-language reasoning to interpret scene context, identify potential anomalies, and localize hazards without requiring additional training on predefined categories. These approaches are specifically designed to address rare and previously unseen scenarios, ensuring that autonomous systems can effectively handle rare and unconventional driving hazards.

III. METHODOLOGIES

Our approach to open-world driving hazard recognition follows a structured, multi-step pipeline with two parallel processes that cross reference each other for efficient hazard detection and captioning in driving scenes. An overview of this process is illustrated in Figure 1. The original COOOL Challenge consisted of three tasks:

- 1) **Hazard Identification:** Selecting the bounding-box-ID of hazards
- 2) **Hazard Captioning:** Generating short captions of the identified Hazard
- 3) **Driver Reaction:** Identifying when the driver reacts to the hazard in the video scene.

In this research, we will focus only on the second task: Hazard Captioning. This approach aligns with our COOOL benchmark annotations, which are specifically designed for hazard captioning. By tailoring the annotations to the second task, we establish a more standardized evaluation framework, improving performance assessment and enabling an in-depth analysis of the application of VLMs and LLMs in autonomous driving for hazard detection.

A. Track 1

The first track focuses on hazard identification and ranking based on how hazardous the object is to the scene through VLMs and LLMs. Given a driving scene video from the dataset, we extract N frames at equal intervals S , where N and

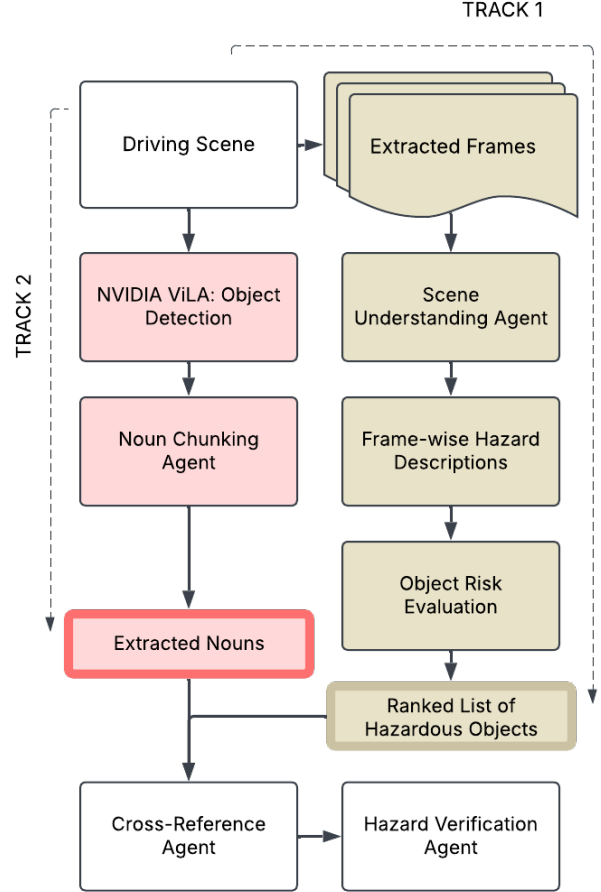


Fig. 1. Our pipeline consists of two parallel tracks, both leveraging vision-language models (VLMs) and large language models (LLMs) to analyze driving scenes. One track focuses on generating detailed linguistic scene descriptions, while the other extracts key object nouns for structured representation. Through iterative refinement, these tracks produce a comprehensive list of elements most likely to be present in the scene. A zero-shot hazard verification agent then evaluates these elements to detect hazardous and anomalous objects with high precision.

S are dependent on the video length to ensure comprehensive temporal representation. For our experiments, we set $N = 25$ for a 10 second video recorded at 30 fps (300 frames total), corresponding to an interval of every $S = 12$ frames. This selection provides sufficient contextual diversity across the clip while maintaining computational efficiency².

For each extracted frame, we query a Scene Understanding Agent, powered by a VLM from one of the following: NexaAI's OmniVLM, NVIDIA's ViLA, or OpenAI's GPT-4o-mini. We used the following prompt for our VLM:

VLM's Prompt: "You are an autonomous vehicle looking to detect hazards or anomalies on the road. A hazard or anomaly could be an animal, person, debris or anything that

²The implications of this decision are discussed in the Future Research section.

is blocking the road. List and describe all the hazards and anomalies in the image.”

This results in N hazardous object descriptions for each video sequence, each detailing potential risks within the scene (e.g. “white van driving at high speed,” “pedestrian crossing unexpectedly,” “stop sign partially obstructed”, etc.). These descriptions are then inputted into a LLM, in our case OpenAI’s GPT-4o-mini, to extract all potentially hazardous objects and to rank them based on what is deemed most hazardous and least hazardous according to the model through the following prompts:

System Prompt: “You are an AI assistant that identifies hazards and anomalies within descriptions of frames from a traffic scene video”

User Prompt: “List1: [List of Descriptions],
List out all the potential hazardous objects and anomalies from most hazardous to least hazardous in the traffic scene and give me very short description of what the object is doing.”³

This then outputs a ranked list which we refer to as *Ranked Hazards Set (RHS)*. This represents the prioritized hazardous elements in the scene.

B. Track 2

The second processing pathway conducts a video wide object detection to comprehensively compile scene elements. Each video is processed using a video understanding agent (ViLA with a temperature of 0.2), which repeatedly queries the video 20 times using the prompt below in order to fully extract all details and objects of the video scene. The low temperature ensures consistent and deterministic responses, preventing unnecessary variation across iterations while enabling thorough extraction of all scene details and objects. We used the following prompt for our VLM in Track 2:

VLM’s Prompt: “What objects are on the road (provide your answer as a list separated by commas)”

This results in 20 sentence outputs, each of which are converted into a python list of the objects detected using the noun-chunking process provided by spaCy [35], a free open source python library used for advanced Natural Language processing such as extracting specific nouns within sentences. These lists are then inputted into OpenAI’s GPT-4o-mini along with the following system and user prompts:

System Prompt: “You are an expert evaluator tasked with selecting the most relevant and comprehensive list from the given options. The best list should contain the most important elements while avoiding redundancy and irrelevant details.”

User Prompt: “Here are multiple lists of objects detected in an egocentric view: [Total list of objects]”
“Please choose the best list based on relevance, completeness, and clarity. The ideal list should include the most important and contextually appropriate elements while avoiding duplicates and unnecessary items. Return only the best list. Also remove any articles (a, the, an) from the list.”

³In this and following prompts, we omit minor prompt details about the format / structure of the output, as this is implementation specific.

The expected output from the LLM is a list of all the detected objects within the video scene (e.g. {“car”, “dog”, “stop sign”, “fire hydrant”, “cyclist”}). We refer to this output as the *All Elements Set (AES)*.

C. Final Hazard Identification and Verification

The final stage of our pipeline intersects these two parallel pathways (Track 1 and Track 2) that output *RHS* and *AES* to refine hazard identification. At this stage, we ask an LLM such as GPT-4o-mini to cross reference the two lists, prompting it with the following:

System Prompt: “You are an AI assistant that cross references two lists.”

User Prompt: “List1: [RHS], List2: [AES]
Cross reference list1 and list2 and output a third list that ranks the common objects as shown in list2. Each index in the list should be a very short sentence description of the object.”

This step filters out hazards that were inferred but not explicitly detected in the scene. The resulting ranked list is referred to as *Critical Object Set (COS)* which contains objects that are both hazardous and present in the driving scene. After *COS* is created, we further query GPT-4o-mini with the following prompt:

System Prompt: “You are an AI assistant that identifies the anomaly such as animals, birds, pedestrians and debris on the road within a list of objects belonging to a traffic scene.”

User Prompt: “List of Traffic Objects: [COS]
Of the listed traffic objects and their descriptions, what are the anomalies / anomaly?”

The resulting is a set that lists anywhere from 1 or multiple identified anomalous hazards and their descriptions in case of a traffic scene containing multiple objects that pose a threat to the traffic scene. We call this set the *Anomalous Object Set (AOS)*. To finalize hazard verification, we utilized OpenAI’s CLIP model alongside the available bounding box data provided by COOOL. For each frame in every video, we extract *snippets*, image regions corresponding to specific areas on the screen, as defined by the bounding box data. However, to ensure meaningful comparisons, snippets with insufficient resolution (i.e., width < 175 pixels, height < 175 pixels, or area $\leq 35,000$ pixels²) are ignored to prevent unreliable matches due to low quality visual data. Each valid snippet is then compared against the descriptive labels in the *COS*. Running these comparisons through CLIP, we compute a similarity score for every snippet-label pair, producing a 2D matrix where each row represents a challenge object (extracted snippet) and each column corresponds to a *COS* label.

The similarity scores range between 0 and 1, indicating the degree of correlation between the extracted visual snippet and its associated textual description. To determine object detection, we select the top 10th percentile of similarity scores in each row: if an element falls within this top percentile, it is marked as detected. Among these detected objects, we cross reference with the Anomalous Hazard Set to identify specific snippets that correlate with marked hazards. This process allows us to track which detected objects are recognized

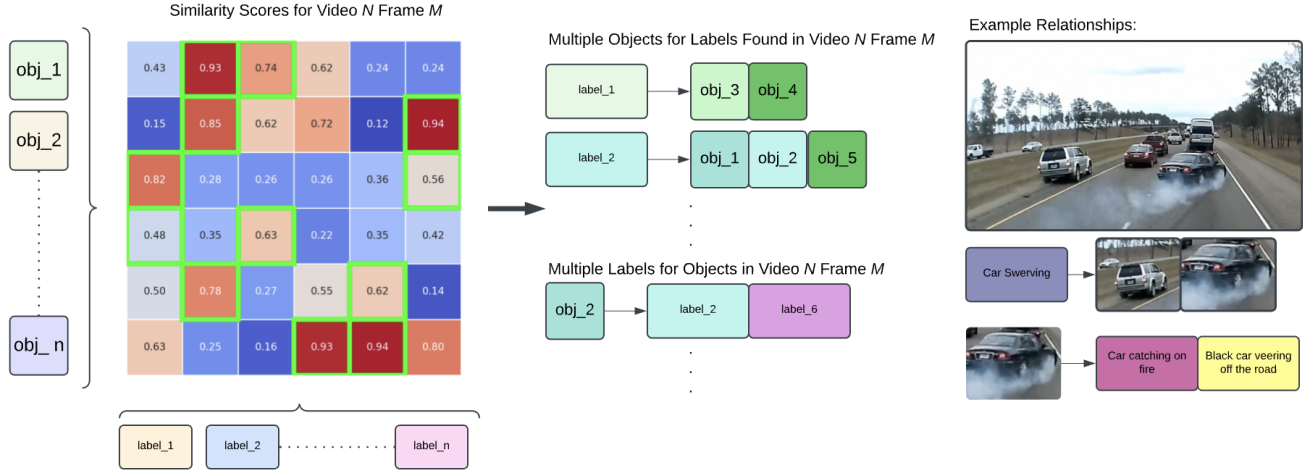


Fig. 2. This figure represents a visualization of our object detection process, where object snippets are compared to label embeddings using OpenAI’s CLIP model. The heatmap displays similarity scores, with each row representing an object snippet and each column corresponding to a potential label. The top 10th percentile of similarity scores (highlighted in green) is selected to identify the most relevant label associations. In the final stage, flagged elements that share common labels are grouped together, enabling more accurate object detection and classification.

as hazardous and the specific frames in which they appear, ensuring precise localization of anomalous hazards within the scene.

IV. EXPERIMENTS AND RESULTS

In our implementation of the above pipeline, we evaluated OmniVLM, ViLA, and GPT-4o-mini as the VLMs utilized in Track 1, used ViLA for Track 2, and selected OpenAI’s GPT-4o-mini as our primary LLM. We ran three experiments, one for each VLM used in Track 1, after preprocessing and annotating the COOOL dataset to create COOLER for performance evaluation.

A. COOLER (Challenge Out-Of-Label with Expanded Representations)

The Challenge of Out-of-Label with Expanded Representations (COOLER) dataset extension builds upon the COOOL benchmark, aiming to enhance the quality and effectiveness of hazard detection in autonomous driving systems and providing new methods of model evaluation. COOLER utilizes the 200 short video clips of COOOL [11], which features various traffic hazards and anomalies, but updates the videos through removal of watermarks and video denoising and deblurring using NAFNet [34]. It also includes a method for open-set evaluation, which includes hazard descriptions via freeform human annotations for the hazardous objects which are then assessed through cosine similarity and embedding based caption analysis. In addition to open-set annotation, our enhancements to input videos included:

- 1) **Watermark Removal** – We manually removed watermarks from the videos using an online watermark remover⁴, ensuring that artificial distractions do not interfere with hazard detection.

⁴<https://online-video-cutter.com/remove-logo>

- 2) **Denoising through NAFNet** – We first deblurred and denoised each frame of a video using NAFNet [12] to remove sensor noise and compression artifacts, enhancing the clarity of key traffic scene features. This improves Vision-Language Models (VLMs) by ensuring cleaner inputs, leading to more accurate feature extraction and scene understanding.
- 3) **Deblurring through NAFNet** – Next, we applied deblurring to restore sharpness and reduce motion blur caused by vehicle movement or camera shake. This enhances fine details, allowing VLMs to extract more precise visual features, improving object recognition and spatial understanding in traffic environments.

By following this procedure, we expect to enhance visual quality, reducing artifacts that could mislead VLMs.

B. Evaluation Methods

This benchmark aims to address hazard captioning tasks through the use of cosine similarities and caption based embeddings. We had human annotators provide two key annotations for each of the 200 video scenes:

- 1) **Hazardous Object Label** – A concise tag identifying the hazard, such as “Cat”
- 2) **Hazard Description** – A short caption describing the event, like “A cat crossing the road”

These annotations serve as reference points for evaluating model performance, ensuring a clear standard for what constitutes a correct detection. To maximize generalization, we designed captions and labels to be as flexible and inclusive as possible (open-set), accounting for variations in phrasing.

Our pipeline produces a Hazard Description A , and we take its cosine similarity with the human-annotated description of the Hazardous Object B to produce a result between 0 and 1 for each video i :

$$S_i = \cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|} \quad (1)$$

We used cosine similarity to evaluate the semantic closeness between predicted and ground truth captions, ensuring models are judged on understanding rather than exact wording. Unlike string matching metrics, cosine similarity recognizes semantically equivalent descriptions, reducing penalization for minor phrasing differences. This improves benchmarking by providing a more accurate assessment of a model's ability to describe hazards, ensuring evaluation reflects true comprehension rather than adherence to predefined labels.

Our benchmark defines a successful detection as a cosine similarity score above 0.80 for Hazard Descriptions. This threshold ensures that predicted hazard labels and descriptions closely match human annotations in meaning, even if phrasing differs; we find that this threshold indicates strong semantic alignment, meaning the model accurately captures the essential information about the hazard.

C. BESM: Balanced Extremes Similarity Metrics

The Balanced Extremes Similarity Metric (BESM) will take the average of the maximum and minimum similarity scores within the Hazard Descriptions for each video i in order to penalize having multiple outputs that may be incorrect within the categories. We then take the average similarity scores for each video over the entire dataset in order to get the BESM of the pipelines performance with the COOLER Benchmark:

For video i :

$$\bar{S}_i^{\text{desc}} = \begin{cases} \frac{\max(S_i^{\text{desc}}) + \min(S_i^{\text{desc}})}{2} & \text{if } |S_i^{\text{desc}}| > 1 \\ S_i^{\text{desc}} & \text{if only one unique value} \end{cases}$$

$$BESM = \frac{1}{N} \sum_{i=1}^N \bar{S}_i^{\text{desc}}$$

D. SAM: Similarity Average Metric

The Similarity Average Metric (SAM) is a straightforward evaluation method that computes an average similarity score across multiple components, ensuring that all similarity values contribute proportionally to the final metric. SAM aggregates description similarity scores, normalizing them based on their respective counts. This approach provides a comprehensive measure that captures variations in similarity across different aspects of the evaluation, avoiding artificial score inflation or arbitrary thresholding. SAM is expressed as:

$$\bar{S}_i^{\text{desc}} = \frac{\sum_{j=1}^m S_{ij}^{\text{desc}}}{m} \quad (2)$$

$$SAM = \frac{1}{N} \sum_{i=1}^N \bar{S}_i^{\text{desc}} \quad (3)$$

where S_{ij}^{desc} are the description similarity scores (total m scores).

TABLE I
FINAL RESULTS OF VLMS ON COOLER

VLM	BESM	SAM
OmniVLM	0.3922	0.3922
ViLA	0.4469	0.4468
GPT-4o-mini	0.3968	0.3967

A key strength of SAM is that it preserves granularity in similarity scores, allowing for a more nuanced representation of similarity relationships. Unlike methods that impose hard cutoffs, potentially exaggerating high scores while disregarding small but meaningful variations, SAM ensures that both low and moderate similarity values contribute meaningfully to the overall evaluation. However, in some cases, SAM may overemphasize superficial word similarities, even when these matches are not semantically significant. This can lead to an unintended bias toward surface-level matches, potentially reducing its effectiveness in prioritizing deeper, more contextually relevant relationships. Given these characteristics, SAM provides a balanced yet sensitive metric that highlights all levels of similarity while maintaining fairness in representation.

Results for both metrics on the COOLER dataset are provided in Table I.

V. DISCUSSION AND ANALYSIS

A. Strengths and Limitations of the Pipeline

This multi-stage validation process, combining linguistic reasoning through VLMS and visual verification through CLIP, produces accurate hazard detection with semantic descriptions. We were able to achieve a scores above 0.39 on both BESM and SAM for each of the three VLMS we evaluated. The models exhibited strong contextual understanding and reasoning, recognizing hazards beyond simple object detection. Additionally, cross referencing and ranking mechanisms refined outputs by aligning results from both tracks, filtering inconsistencies, and prioritizing critical hazards, making the system highly applicable to real-world autonomous driving.

Though evaluating on a different set of metrics, at the time of writing, the initial COOL challenge leaderboard has an average score among top 15 of 0.386 [36] with the best score being 0.573 [37]. We do not intend to make any direct comparison here, but simply to state that anomaly detection on this dataset is a challenging task; the best models (even on different metrics) are not achieving “near-perfect” performance, leaving significant room for future research.

The pipeline fails to detect small hazards within some of the videos, but when it does detect hazards, it does so with good semantic performance. A key observation from our experiments is that there were multiple instances where the pipeline generated no output which hurt the scores for BESM and SAM. This primarily occurred due to two reasons:

Failure to Detect Hazardous Objects: There is sometimes failure in the merging of Track 1 and Track 2 outputs, where the VLMS or LLM did not detect a hazardous object. This

fault could have happened for two reasons: (1) the VLMs were not able to detect the hazardous object within the video sequence, or (2) the LLM was not able to reason for any of the listed hazardous objects to be hazardous enough.

Object Size Limitations: Some hazards were not detected because the bounding box annotations of the hazardous objects were too small, leading to their exclusion from our pipeline. These limitations highlight the need for refining bounding box processing within our pipeline in order to pass through smaller objects.

B. Evaluation of COOLER

The COOLER Benchmark provides a structured evaluation of hazard detection capabilities through multiple metrics. BESM penalizes inconsistent outputs, revealing stability issues but being sensitive to outliers. SAM provides a balanced average similarity but struggles to highlight peak performance. Our evaluations also exposed limitations, such as failures in detecting small or occluded hazards and score fluctuations due to minor linguistic variations. Additionally, the driving scenes in our dataset occasionally featured scenarios where no true hazard was present or where the hazard was insignificant to the driver. For example, one scene captures a driver on the freeway passing an exit ramp, where a small piece of debris is present on the ground. While technically an object in the environment, such instances do not pose a direct threat to the driver. Our pipeline is designed to intentionally ignore hazard objects that are too small to be meaningfully detected, prioritizing actionable hazards over minor environmental anomalies. However, this results in penalization under the COOLER evaluation metric, as the final scores are averaged across all videos, including those where small, non-threatening objects are present but not flagged by the system. This highlights a tradeoff between filtering out inconsequential hazards and maintaining high recall in benchmark evaluations. Despite these challenges, COOLER effectively identified strengths and weaknesses in our pipeline, emphasizing the need for improvements in bounding box processing, scene context modeling, and evaluation flexibility to enhance detection accuracy and robustness.

VI. CONCLUDING REMARKS AND FUTURE RESEARCH

While our pipeline significantly improves hazard detection accuracy, several challenges remain. Failures in merging detection tracks and difficulties recognizing small or occluded objects impacted performance. Future research should refine bounding box processing, cross referencing strategies, and motion based reasoning to enhance object recognition and reduce false negatives. Additionally, further prompt engineering to increase the contextual understandings of VLMs and LLMs could also be quite beneficial.

While our use of 25 evenly spaced frames per video ensured broad temporal coverage and significantly reduced computational overhead, this sparse sampling may miss fast moving or short duration hazards that occur between frames. Additionally, it limits the model's ability to capture temporal

dynamics such as the appearance, evolution, and dissipation of hazards, key elements for accurate reasoning in real-world driving scenarios. Although consistent sampling aids standardization across clips, future work should explore processing all frames or adopting dense temporal sampling to better preserve motion cues and improve fine grained hazard detection and prediction.

Our pipeline successfully leveraged OmniVLM, ViLA, GPT-4o-mini as our VLM and GPT-4o-mini as our LLM, further exploration of alternate combinations of different VLMs and LLMs could help with the evaluation of detection accuracy and reasoning capabilities between models. Different combinations of models can be used to evaluate their reasoning and to also help address the limitations where the pipeline fails to reason for and detect hazardous objects in the COOLER dataset's video sequences. Exploration of the combinations of different models could also be beneficial in evaluating their applicability within autonomous driving scenes.

This research has focused on hazard captioning, but the original COOL Benchmark also includes the identification of the bounding box labels of the hazardous object and Driver Reactions task determining when a driver responds to a hazard. Future work should explore temporal modeling approaches such as transformers or recurrent neural networks to track hazard evolution and predict driver reactions. Integrating sequential reasoning would enhance our ability to model cause-and-effect relationships within dynamic driving scenarios, further improving safety assessment in autonomous systems.

This research advances hazard detection in autonomous driving by integrating the zero-shot reasoning of VLMs and LLMs for the open-set recognition of novel hazards beyond predefined categories. Through the COOLER Benchmark, we established a structured evaluation framework, enhancing dataset quality with video preprocessing, refined annotations, and new methods for evaluation on hazard descriptions provided by the pipeline's output. Our multi-agent pipeline, demonstrated strong context aware hazard identification, improving adaptability and detection reliability, but limitations still remained particularly in handling small or occluded hazards and refining detection consistency across frames. This work lays the foundation for real-time hazard assessment, predictive autonomous systems, and safer AI driven navigation. As vision-language models continue to evolve, their integration into autonomous driving has the potential to significantly enhance traffic safety, reduce accident risks, and improve situational awareness, bringing autonomous vehicles closer to human level hazard perception.

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